ECE 447

Lesson 22
Digital Communication and Line Coding

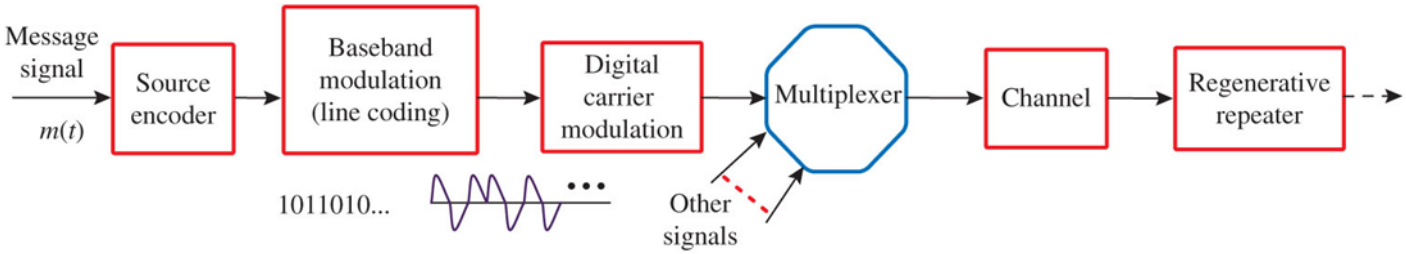


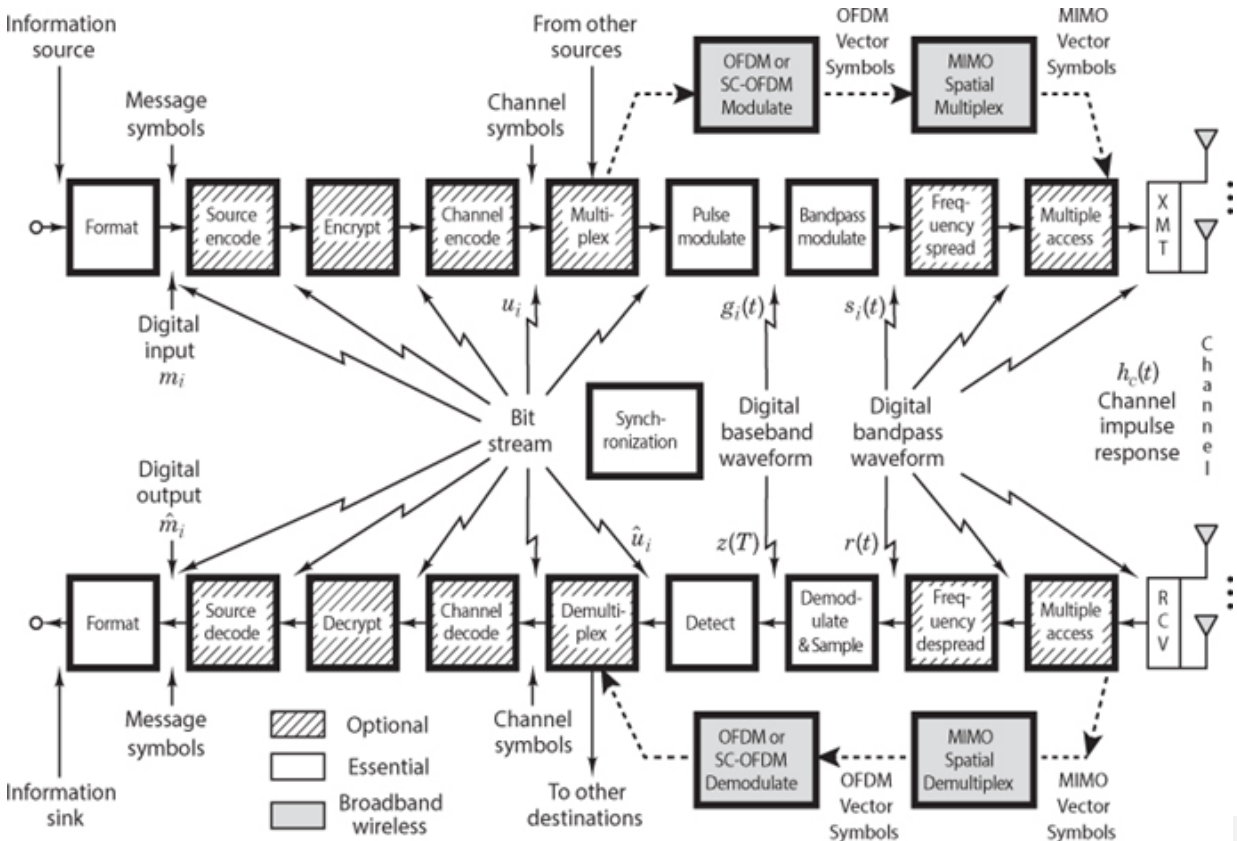
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Schedule and Admin

- Schedule
- Admin
  - **HW4**. Was due Lesson 21 (6 Oct).
  - **HW5**. Due Lesson 23 (Tue, 13 Oct) to Gradescope.
  - **Lab 5**. Due Lesson 27 (Fri, 23 Oct) to Gradescope.

Digital Communication Systems



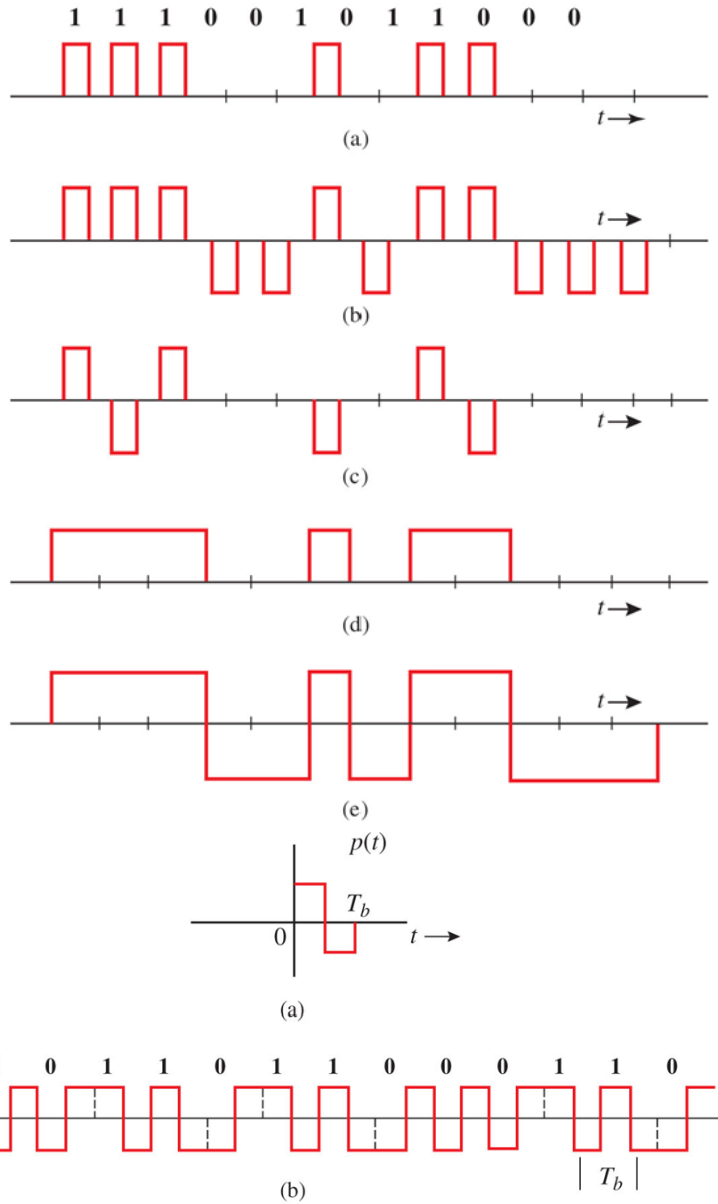


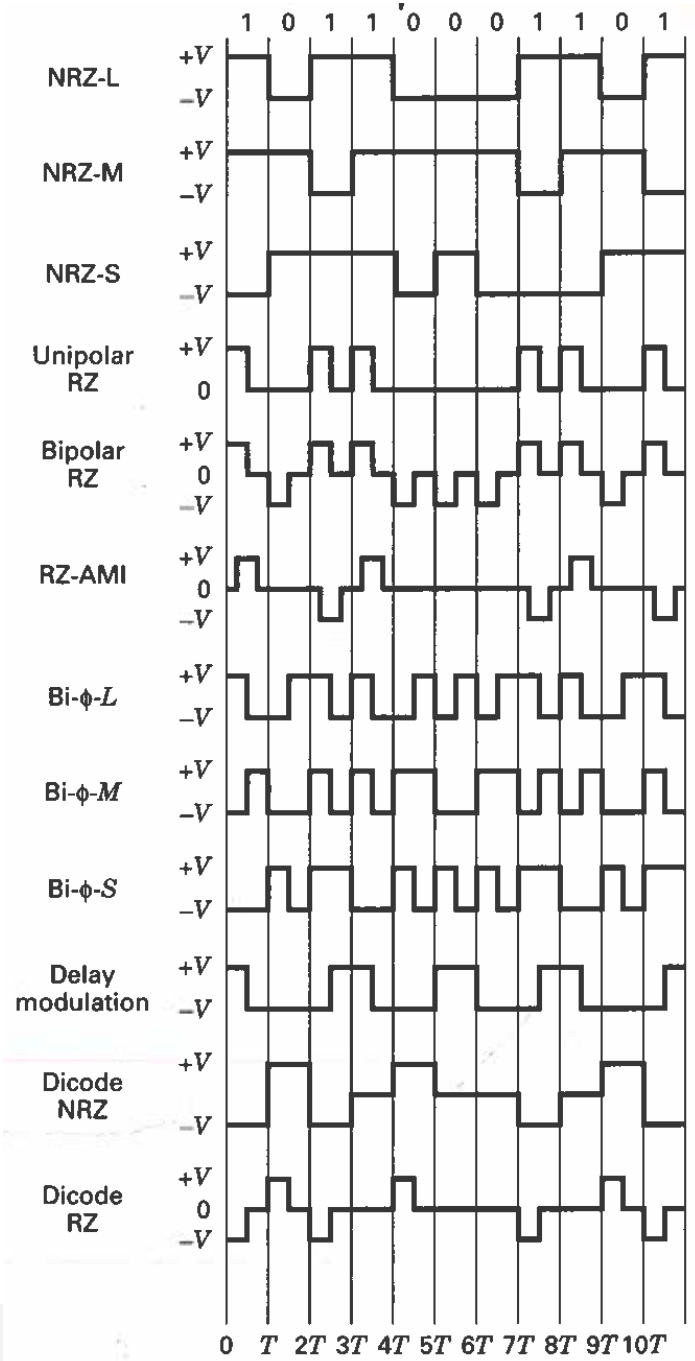
Baseband Line Coding

Desired line code properties:

- Low bandwidth
- Power efficiency - low tx power for given bandwidth and error detection rate
- Error detection/correction capability
- No DC content
- Timing info (self-clocking)
- Transparency

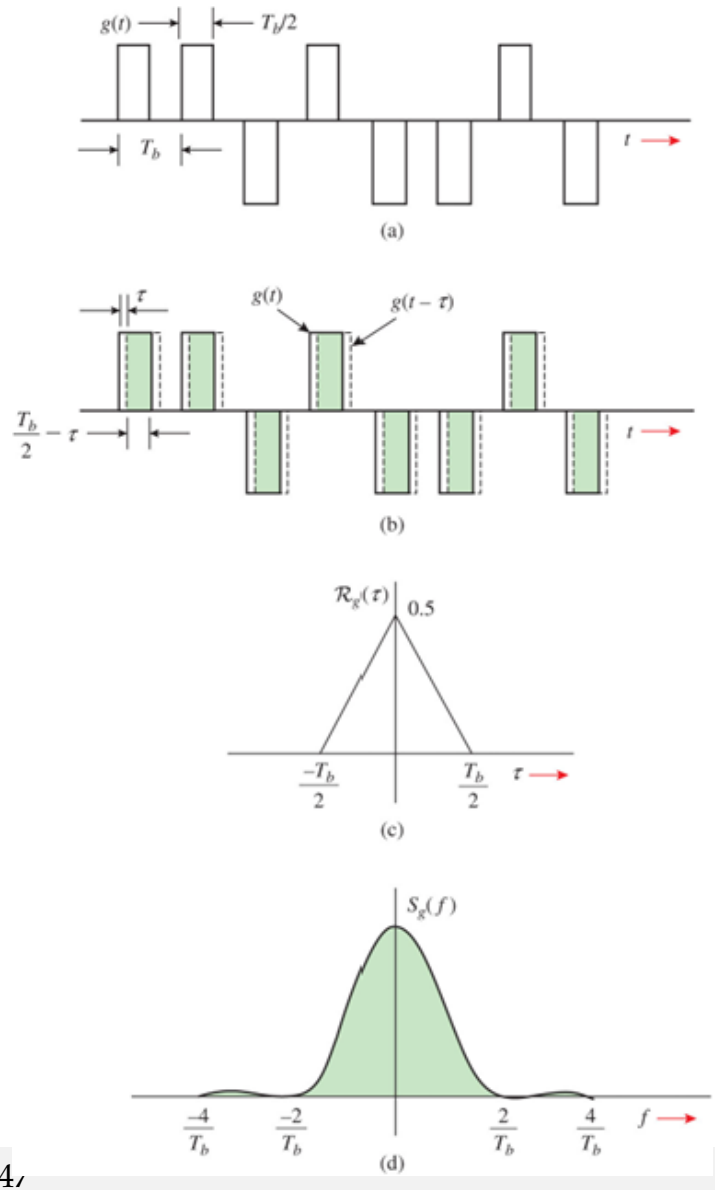
Types of Line Codes



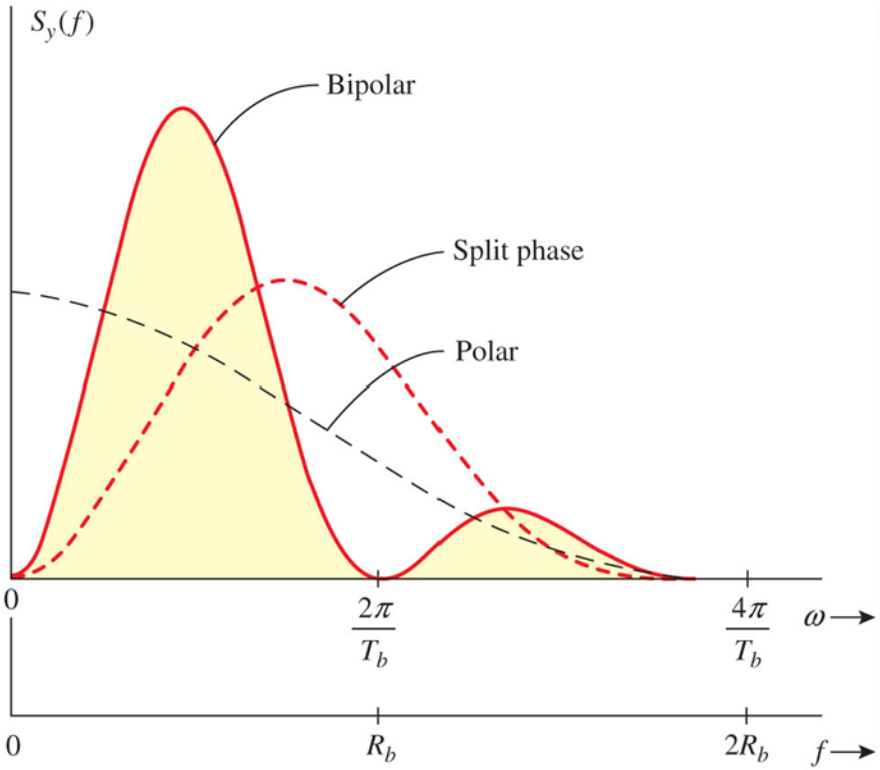


Line Code PSDs

Example 3.24 - Our good friend, Autocorrelation!



Line Code Spectra



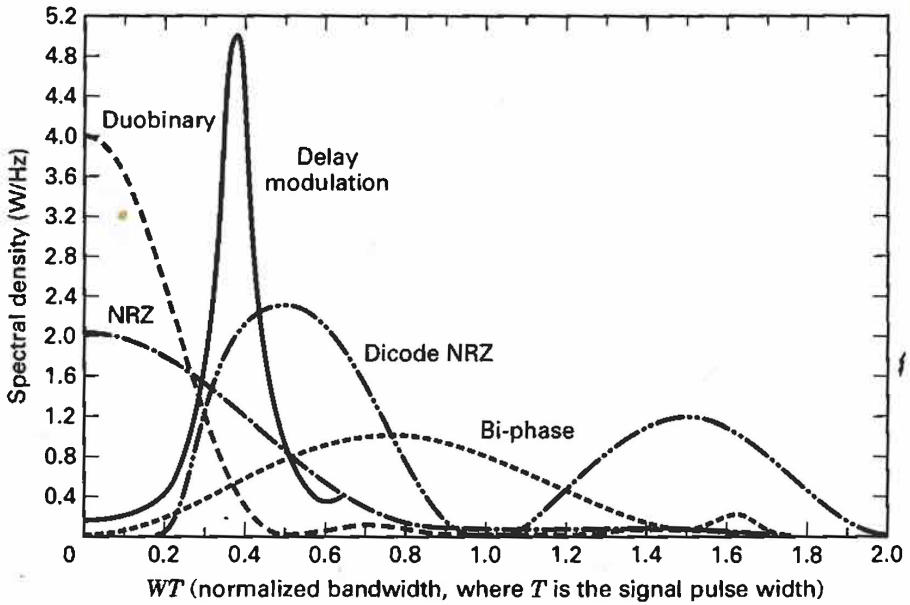


Figure 2.23 Spectral densities of various PCM waveforms.

Example: Change the Pulse, Keep the Code

- The pulse shape and the line code enter the PSD **separately**: $S_y(f) = \frac{|P(f)|^2}{T_b} \times$ (code factor)
  - Polar: 1 On-off: $\frac{1}{4}\left[1 + \frac{1}{T_b}\sum_{n=-\infty}^{\infty}\delta\left(f - \frac{n}{T_b}\right)\right]$ Bipolar: $\sin^2(\pi f T_b)$
  - Change the pulse and **only** $|P(f)|^2$ moves — the code factor stays put
- **Try it.** Polar signaling with a *half-width* RZ pulse, $p(t) = \Pi\left(\frac{2t}{T_b}\right)$. You need $\Pi(t/\tau) \leftrightarrow \tau\,\mathrm{sinc}(\pi f \tau)$.
  - What is τ here? _____ So $P(f) =$ _____
  - $|P(f)|^2 =$ _____ $S_y(f) =$ _____
  - First null where $\pi f T_b/2 = \pi$, so $f =$ _____
- Full-width NRZ, $\Pi(t/T_b)$, nulls at R_b. Halving the pulse width _____ the bandwidth.
- HW6 Problem 1 runs this same recipe on a different pulse: find the pair, square it, multiply.

Line Code to Baseband Waveform

